Where the Shortcut Sits

Status report 1: does masking the region of interest remove artifact shortcuts?

October 2026

Summary.

- Medical image classifiers often learn *artifacts* (rulers, hair, calipers, debris) instead of the disease. The usual defence is *ROI masking*: keep only the region of interest (ROI), such as the lesion, and delete the rest of the image.
- Masking removes the artifact only when the artifact lies *outside* the ROI. When it lies *inside*, masking keeps it and deletes everything else, and the model relies on the artifact *even more*.
- We showed this in four steps: (1) moving a pasted ruler into a skin lesion step by step; (2) four checks of *why* it happens; (3) the same test in four other kinds of medical images; (4) real, unedited artifacts in four patient cohorts.
- With real artifacts, masking helped much more when the artifact was outside the ROI than when it was inside, in all four cohorts (all four planned tests significant after correction for multiple testing). Two image encoders were exceptions (Section 7).

1 The problem

A deep classifier learns whatever predicts the label in its training data, including things that are not the disease (Geirhos et al., 2020). This is called a *shortcut*. In medical images the shortcut is often an artifact that clinicians put there because they suspected disease: a ruler or skin marking next to a suspicious mole (Winkler et al., 2019, 2021), calipers on a thyroid nodule that was measured, or a chest drain in a patient with a collapsed lung (Jiménez-Sánchez et al., 2023). Because these artifacts appear more often in sick patients, a model can score well by detecting the artifact. It then fails on exactly the patients whose images look different, for example a melanoma photographed without a ruler.

The most common defence is to segment the lesion and give the model only that region; the rest of the image is deleted. This is called *region-of-interest (ROI) masking*. It is popular because it needs only a lesion mask, which many pipelines already compute. Its hidden assumption is that the artifact lies in the background. Rulers, hair, calipers and debris often lie on or across the lesion. Figure 1 shows what masking then does.

Research question. *Does the benefit of ROI masking depend on where the artifact sits relative to the ROI? If so, why, and does it hold for real artifacts and other kinds of medical images?*

2 How we test it

Two models Every experiment compares two models that use the same frozen image encoder (a pretrained foundation model such as DINOv2 (Oquab et al., 2024)) and the same simple classifier on top:

- the **standard model** sees the whole image;
- the **masked model** sees the image with everything outside the ROI replaced by the image's average colour, in training and in testing.

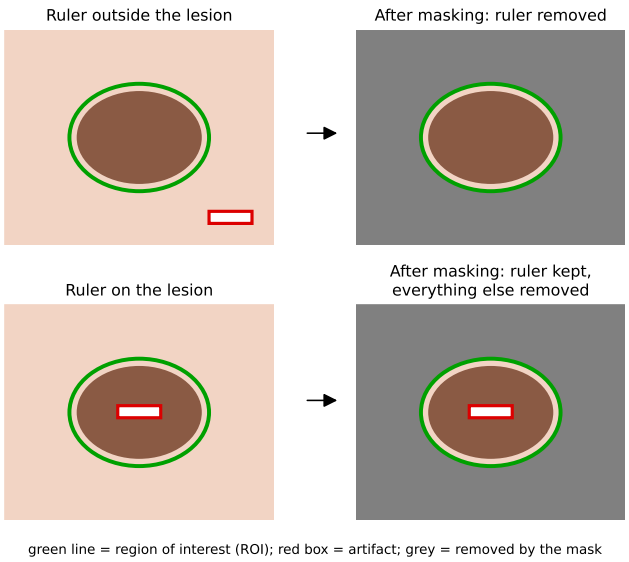


Figure 1: The idea. Top: an artifact outside the lesion is deleted by the mask. Bottom: an artifact on the lesion survives the mask, while everything that competed with it is deleted.

We keep the encoder frozen so that the only difference between the two models is the input.

The trap To measure how much a model relies on the artifact, we build a "trap" (Figure 2). In the training data the artifact appears on 90% of the diseased images and 10% of the healthy ones, so it is a strong shortcut. We then test on three test sets:

- **correlated test**: same 90/10 pattern as in training. A shortcut model looks good here.
- **reversed test**: the pattern is flipped (10/90). A model that uses the artifact *fails* here.
- **clean test**: the artifact is unrelated to the disease. This shows how well the model knows the disease itself.

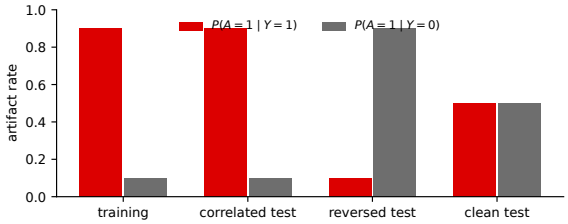


Figure 2: The trap. Bars show how often the artifact appears among diseased (red) and healthy (grey) images in training and in each test set.

The score We report AUROC (0.5 = guessing, 1.0 = perfect). Our main score is the AUROC on the *reversed* test: it is high only if the model looks at the disease and not at the artifact. We usually report the *difference* "masked minus standard". A positive number means masking helped; a negative number means it hurt. Square brackets give the 95% confidence interval; if it does not include zero, the difference is significant. The **location effect** is this difference with the artifact outside the ROI minus the same difference with the artifact fully inside. A positive location effect means masking helps more when the artifact is outside.

Measuring fairly Two details turned out to matter.

- **Near-duplicate images.** Skin-lesion datasets contain several photos of the same lesion. If one copy is in training and another in testing, the test partly measures memory. We always split by

lesion (and by image similarity), never by single images. With random splits, 93 near-duplicate pairs crossed the split and the measured shortcut was 10–18% too large (Table 1).

- **Honest error bars.** We train each model several times (different random seeds), but all runs are tested on the *same* test images. A common way of computing confidence intervals treats these runs as if each had its own test set, which makes the intervals too narrow. We resample the seeds and the test images together, so the test set counts only once. Our intervals are about 1.5 times wider, and a few results that looked significant with the common method are not.

Before each main experiment we wrote down the design and what result would count as support, and stored it in the project repository. Results are reported whichever way they went.

Table 1: Why we split by lesion: random image-level splits leak near-duplicates and inflate the shortcut (ruler experiment of Section 3, DINOv2). "Shortcut gap" = correlated minus reversed AUROC of the standard model.

Ruler position	Split	Near-duplicate pairs across the split	Shortcut gap	Masked − standard
outside the lesion	by lesion	0	0.241	+0.184
	random (mean of 5)	93	0.284	+0.223
on the lesion	by lesion	0	0.421	−0.129
	random (mean of 5)	93	0.465	−0.088

3 Step 1: moving a ruler into the lesion

What we did We first wanted to change *only* the location of the artifact. We took 2,437 dermoscopy images with manual lesion masks (ISIC 2018 (Codella et al., 2019); task: melanoma or not) and pasted the same ruler into each image at 0, 25, 50, 75 or 100% overlap with the lesion (Figure 3). Only the ruler's position changes: same images, same ruler, same labels.

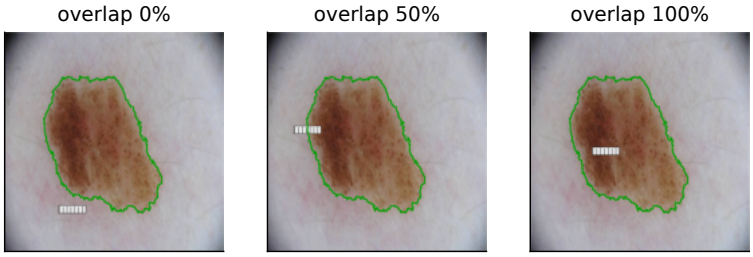


Figure 3: The same image with the ruler at 0, 50 and 100% overlap with the lesion (green: lesion mask).

What we found When the ruler lies outside the lesion, masking removes it and helps a lot (+0.18). From 50% overlap on, masking *hurts*: at full overlap the masked model is worse than the standard model (−0.13; Table 2, Figure 4). The correlated test shows why: with the ruler on the lesion, the masked model scores *higher* than the standard model there (0.961 vs 0.937) but lower on the reversed test (0.386 vs 0.515). It has become *more* dependent on the ruler.

Table 2: Ruler experiment (DINOv2, AUROC, mean of 3 seeds). The last row is the effect of masking on the reversed test.

	Ruler outside (0%)	Half on the lesion (50%)	Fully on the lesion (100%)
Standard model, reversed test	0.665	0.573	0.515
Masked model, reversed test	0.849	0.476	0.386
Standard model, correlated test	0.906	—	0.937
Masked model, correlated test	0.849	—	0.961
Masked − standard (reversed)	**+0.184** [+0.117, +0.250]	**−0.097** [−0.164, −0.030]	**−0.129** [−0.195, −0.064]

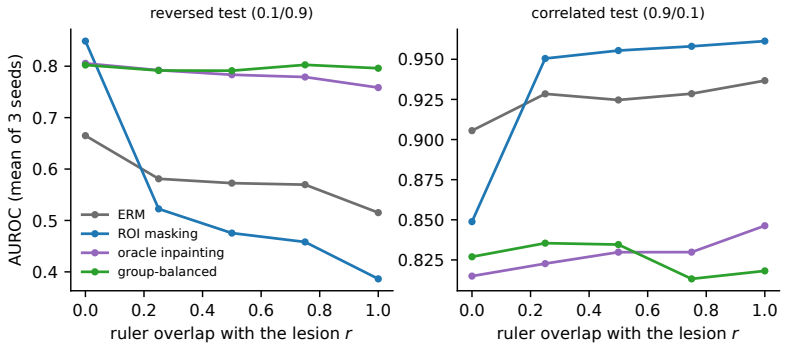


Figure 4: AUROC on the reversed test (left) and the correlated test (right) as the ruler moves into the lesion. Blue: masked model; grey: standard model. For comparison: purple, painting out the ruler (needs the ruler's outline); green, balanced reweighting (needs to know which images have a ruler).

Is it a fluke of one model or one ruler? No. We repeated the experiment with a lower image resolution (224 px), with a dermatology-specific encoder (DermLIP (Yan et al., 2025)), and with a ruler whose colour, ticks and size change from image to image. The location effect was large and positive every time (Table 3, first four rows).

Table 3: The ruler result repeated with other settings (masked − standard, reversed test).

Setting	Ruler outside	Ruler fully inside	Location effect
DINOv2, 518-px images (main result)	+0.184	−0.129	+0.313 [+0.258, +0.370]
DINOv2, 224-px images	+0.361	−0.095	+0.456 [+0.390, +0.525]
DermLIP (dermatology encoder)	+0.161	−0.263	+0.424 [+0.335, +0.517]
DINOv2, randomised ruler	+0.146	−0.155	+0.302 [+0.248, +0.357]
Two ruler sizes on the same 1,314 images:			
DINOv2, ruler 13% of the image width	+0.160	−0.102	+0.262 [+0.200, +0.325]
DINOv2, ruler 39% of the image width	+0.170	−0.054	+0.224 [+0.161, +0.285]
Average of the two sizes	**+0.165**	**−0.078**	**+0.243** [+0.182, +0.304]

Different ruler sizes Real rulers come in different sizes. We therefore ran two sizes, 13% and 39% of the image width (the wider one thin, with millimetre ticks, like a real scale bar), on the 1,314 images whose lesions can hold both fully inside (last three rows of Table 3). The design was written down before the run; the average was added afterwards. Both sizes show a clear location effect (+0.262 and +0.224). Averaged over the two, masking helps by +0.165 with the ruler outside and hurts by −0.078 [−0.149, −0.010] with it fully on the lesion. For the 39% ruler alone, that harm is not significant (−0.054; the interval includes zero). With both sizes, the masked model again reacts to the ruler on the lesion about twice as strongly as the standard model (check 3 below).

4 Step 2: why does masking fail?

Step 1 showed *that* masking fails when the ruler is inside. There were three possible reasons: (a) masking simply hides useful parts of the image; (b) the background holds disease information that masking throws away; (c) the ruler inside the lesion survives, and because masking deletes everything that competed with it, the model relies on it more. We designed one check for each (Table 4).

Explanation. Masking deletes everything outside the ROI. If the artifact is outside, the shortcut is gone. If the artifact is inside, it survives, and the other image information that used to compete with it is gone, so the artifact becomes a bigger part of what the model can use. The model relies on it more, and fails when the artifact is missing or misleading.

Table 4: Four checks of why masking fails (ruler experiment, DINOv2).

Check	What we changed	Result	What it means
1. Useless ruler	Ruler at the same places, but *unrelated* to the diagnosis	Masking does not hurt at any position (all intervals include 0; e.g. +0.039 at 100%)	Hiding parts of the image is not the problem (rules out a)
2. Bigger mask	Grow the mask around the lesion: more skin comes back, and more of a half-covered ruler stays visible (50% → 95% of the ruler)	Harm gets *worse*: −0.097 → −0.153. When the ruler is already fully visible, a bigger mask changes nothing (−0.129 vs −0.138)	Harm follows how much of the ruler is visible, not how much skin is lost (b is not the main cause)
3. Same-image test	Add or remove the ruler on the *same* test image and measure how much the prediction changes	The masked model's prediction changes about twice as much as the standard model's (0.49 vs 0.24; with the 39% ruler 0.43 vs 0.21)	The masked model leans more on the ruler (supports c)
4. Lesion size	Split test images into small, medium and large lesions	Harm with the ruler inside: small −0.36, medium −0.16, large +0.06 (not significant)	Harm is largest where the ruler is big compared to the lesion

5 Step 3: is it only skin images?

What we did We repeated the moving-artifact experiment in four other kinds of medical images, each with its own typical artifact and its own ROI (Figure 5): a sonographer's caliper on **thyroid ultrasound** (TN3K, malignant or benign nodule (Gong et al., 2021)) and on **ovarian ultrasound** (MMOTU, tumour with or without a solid part (Zhao et al., 2022)); a debris blob on **capsule endoscopy** (SEE-AI, erosion or polyp-like lesion (Yokote et al., 2023)); and a tube on **chest X-rays** (NIH ChestX-ray14, pneumothorax (Wang et al., 2017); ROI = the lungs). We used only images without a real artifact, so the pasted one was the only one. We also tested a second encoder where possible (MedSigLIP, a medical image model; RAD-DINO, a chest X-ray model (Pérez-García et al., 2025)).

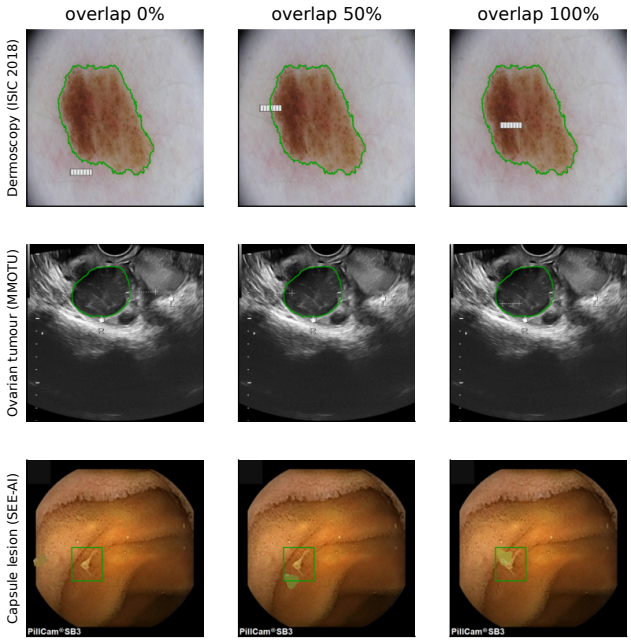


Figure 5: The pasted artifact at 0, 50 and 100% overlap in three modalities (green: ROI; for capsule images the ROI is the expert's box around the lesion).

What we found Counting the first four rows of Table 3 and the seven rows of Table 5, the location effect was positive in 10 of 11 experiments (Figure 6). Masking helped clearly when the artifact was outside and lost most or all of its benefit when it was inside. Whether masking actually *hurt* at full overlap depended on the setting: it hurt significantly in 7 of the 11. In thyroid ultrasound, for example, the benefit fell from +0.40 to about zero.

Table 5: Moving-artifact experiments in other modalities (masked − standard, reversed test).

Modality and artifact	Encoder	Artifact outside	Artifact fully inside	Location effect
Thyroid ultrasound, caliper	DINOv2	+0.402	−0.030	+0.433 [+0.362, +0.507]
	MedSigLIP	+0.514	+0.044	+0.470 [+0.388, +0.556]
Ovarian ultrasound, caliper	DINOv2	+0.041	−0.197	+0.238 [+0.144, +0.340]
Capsule endoscopy, debris	DINOv2	+0.116	−0.112	+0.228 [+0.144, +0.324]
	MedSigLIP	+0.374	+0.216	+0.158 [+0.012, +0.314]
Chest X-ray, tube	DINOv2	+0.254	−0.083	+0.337 [+0.312, +0.364]
	RAD-DINO	−0.002	+0.111	−0.113 [−0.129, −0.097]

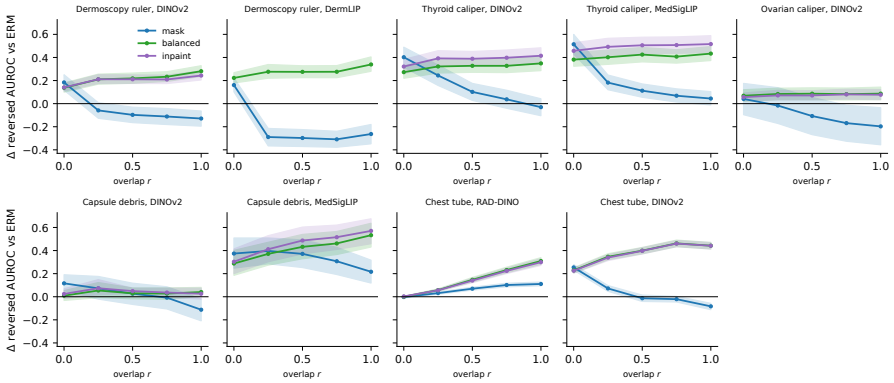


Figure 6: Effect on the reversed test (vs the standard model) as the artifact moves into the ROI, in every moving-artifact experiment. Blue: masking; purple: painting out the artifact; green: balanced reweighting. The blue line falls in every panel except the RAD-DINO chest tube.

The one exception With the chest X-ray encoder RAD-DINO, the standard model ignored a tube outside the lungs (reversed AUROC 0.913, the same as on the clean test, 0.915). Masking had nothing to remove outside, so the comparison does not apply. With DINOv2, which did use the outside tube, the usual pattern appeared.

6 Step 4: real artifacts

Pasted artifacts let us control location exactly, but they are not the artifacts clinicians produce. So we next asked whether the same thing happens with *real, unedited* artifacts.

The design A real artifact cannot be moved, so we compare two groups of images: those whose artifact lies mostly inside the ROI (**Trap A**) and those whose artifact lies outside it (**Trap B**). Both traps share the *same* artifact-free images (Figure 7), so the designed difference between them is where the artifact sits. Each trap gets the 90/10 training set and the three test sets of Figure 2. The **crossover** is the benefit of masking in Trap B minus its benefit in Trap A; a positive crossover means masking helps more when the artifact is outside. (We first tried comparing in-ROI with out-of-ROI images directly, without an artifact-free group. That design cannot work: masking deletes exactly the out-of-ROI artifacts that define one side of the comparison, so it shows "harm" whatever the truth.)

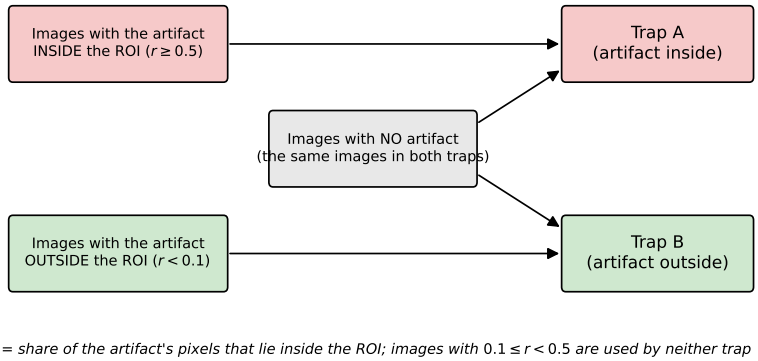


Figure 7: Two traps built from real images. Both share one group of artifact-free images.

Cohorts Table 6. Artifacts were located automatically: published hair masks for skin images (Wegley et al., 2026), a rule-based caliper detector for ultrasound, and a small detector trained on expert debris masks for capsule images. We checked them by eye: the caliper detector was right on 38 of 40 checked images. Detection errors that do not depend on the artifact's location weaken the shortcut in both traps; they cannot by themselves create a crossover.

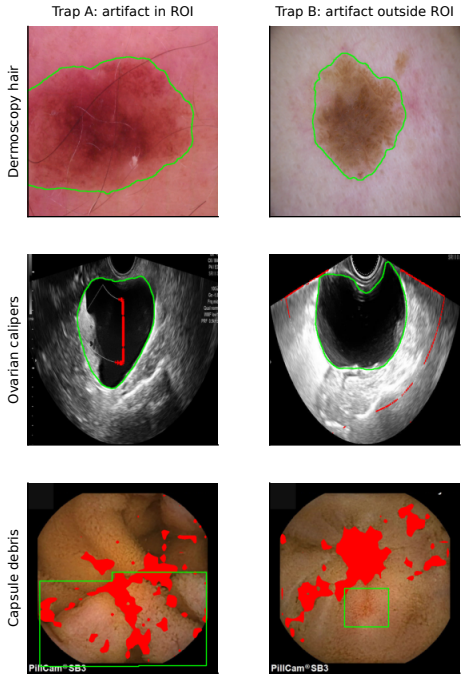


Figure 8: Real artifacts inside (Trap A) and outside (Trap B) the ROI. Green: ROI; red: detected artifact. (Thyroid images are not shown because the data licence is unclear; hair masks are not drawn for licence reasons.)

Table 6: The four real-artifact cohorts (number of images, diseased / not diseased).

Cohort	Task	Artifact	No artifact	Trap A (inside)	Trap B (outside)
Skin lesions, ISIC 2019 (3 hospitals)	melanoma vs other	hair	157 / 706	942 / 2,655	969 / 4,412
Thyroid ultrasound, TN3K	malignant vs benign	calipers	749 / 1,052	292 / 858	99 / 227
Ovarian ultrasound, MMOTU	solid part vs cystic	calipers	178 / 170	276 / 341	103 / 91
Capsule endoscopy, SEE-AI	erosion vs polyp-like	debris, bubbles	343 / 164	412 / 404	1,009 / 231

What we found The crossover was positive in all four cohorts with DINOv2 (Table 7). These four tests were planned in advance, and all four stay significant after Holm correction for testing four cohorts (adjusted $p = 0.0004$ each). It also held with MedSigLIP and with a convolutional network (ConvNeXt), so it is not specific to one type of model; it did not hold with DermLIP (Figure 9, Section 7).

Table 7: Real artifacts: effect of masking on the reversed test (masked − standard) in each trap, and the crossover.

Cohort	Encoder	Trap A (inside)	Trap B (outside)	Crossover (B − A)
Skin, hair	DINOv2	−0.049	+0.098	+0.147 [+0.111, +0.185]
Thyroid, calipers	DINOv2	+0.109	+0.348	+0.239 [+0.196, +0.280]
Ovary, calipers	DINOv2	+0.132	+0.302	+0.170 [+0.087, +0.252]
Capsule, debris	DINOv2	+0.084	+0.461	+0.377 [+0.327, +0.426]
Thyroid, calipers	MedSigLIP	+0.053	+0.248	+0.195 [+0.153, +0.238]
Ovary, calipers	MedSigLIP	+0.002	+0.180	+0.178 [+0.109, +0.248]
Capsule, debris	MedSigLIP	+0.117	+0.452	+0.335 [+0.286, +0.383]
Thyroid, calipers	ConvNeXt	−0.004	+0.271	+0.275 [+0.231, +0.320]
Capsule, debris	ConvNeXt	+0.084	+0.403	+0.320 [+0.268, +0.372]
Skin, hair	DermLIP	−0.104	−0.090	+0.014 [−0.037, +0.063]

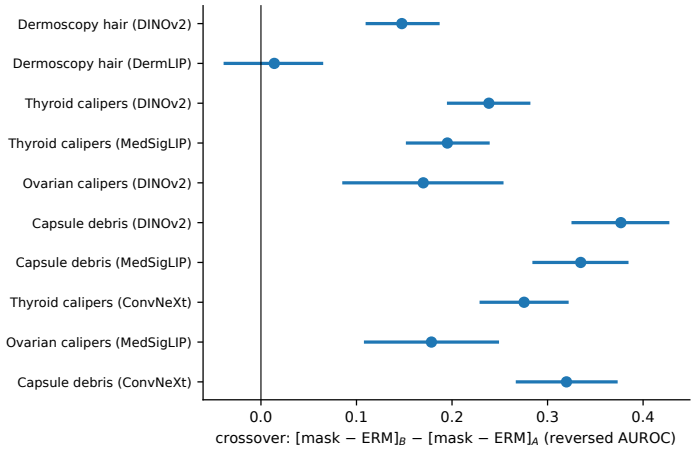


Figure 9: The crossover for every cohort and encoder (dots: estimate; lines: 95% interval). Right of zero = masking is worth more when the artifact is outside the ROI.

How to read it For skin hair, masking *hurt* when the hair lay on the lesion. For calipers and debris, masking still helped in Trap A, because it removes other on-screen marks and clutter, but much less than in Trap B, and a large part of the shortcut stayed. In thyroid Trap A, the masked model scored 0.908 on the correlated test but only 0.398 on the reversed one; in Trap B masking closed this gap (0.785 vs 0.752). The general rule is: *masking helps much less when the artifact is inside the ROI.* Whether it actually hurts depends on how much useful image information masking also removes.

7 Other things we learned

- **Methods that know which images carry the artifact help in both traps.** Reweighting the training images so that the artifact no longer predicts the label, or retraining the last layer on balanced data (DFR (Kirichenko et al., 2023)), helped in both traps of every cohort, wherever

the artifact lay (Table 8). The catch: they need an artifact label for every training image, which clinical datasets rarely have. Masking needs only the ROI.

- **Painting out the artifact can backfire.** In capsule images, painting over the detected debris *lowered* the score (−0.17 in Trap A), because erosions contain yellow-white material that looks like debris. Deleting artifact pixels is only safe when the artifact does not look like the disease.
- **The hair shortcut is not only in the hair pixels.** Hair goes together with patient age, sex and body site. Painting out the hair pixels closed less than half of the standard model's shortcut gap (47% in Trap A, 44% in Trap B); the rest comes from these related factors, which masking cannot remove.
- **DermLIP did not follow the rule.** With this dermatology encoder, masking hurt in *both* hair traps and there was no crossover. It needs the skin around the lesion, so any mask costs disease information.
- **Chest X-rays are a boundary case.** Real chest drains lie inside the lungs and are an extreme shortcut (standard model, RAD-DINO: reversed AUROC 0.19 against 0.89 on the clean test). Lung masking barely helped (0.23), while DFR reached 0.67. With real devices (catheters in the lungs vs breathing tubes outside them), RAD-DINO showed no crossover and MedSigLIP a positive one in all four findings tested. The two device groups also differ in which other devices are present, so we do not count chest X-rays as a confirmation.

Table 8: Masking against methods that use artifact labels (reversed test, change vs the standard model, DINOv2).

	Skin, hair		Thyroid, calipers		Capsule, debris	
Method	Trap A	Trap B	Trap A	Trap B	Trap A	Trap B
ROI masking (needs ROI only)	−0.049	+0.098	+0.109	+0.348	+0.084	+0.461
Balanced reweighting (needs labels)	+0.213	+0.204	+0.388	+0.254	+0.179	+0.244
DFR (needs labels)	+0.196	+0.196	+0.463	+0.312	+0.234	+0.287
Painting out the artifact (needs its mask)	+0.098	+0.100	+0.102	+0.039	−0.172	−0.115

8 Where we are

Table 9: What the work so far shows.

Question	Answer	Evidence
Does masking depend on where the artifact sits?	Yes: it helps when the artifact is outside the ROI and loses its benefit (often turns harmful) when it is inside	Steps 1 and 3: 10 of 11 experiments; both ruler sizes
Why?	The artifact inside survives, its competitors are deleted, and the model relies on it more	Step 2: four checks
Does it hold for real artifacts?	Yes, in all four cohorts and with three encoder types	Step 4: four planned tests, all significant
Exceptions	DermLIP (needs the skin around the lesion); RAD-DINO (ignores the artifact outside the lungs)	Steps 3 and 4

Questions for discussion

1. Is the reversed-test AUROC acceptable as the main score, or should a clinical measure (sensitivity at a fixed specificity) be added from the start?

2. Should the rule be stated as "masking is worth less inside the ROI" (true in all real cohorts) rather than "masking hurts inside the ROI" (true for skin hair and in 7 of 11 moving-artifact experiments)?
3. The artifact masks were checked by eye by a non-clinician. Should a clinician check a sample?
4. How should chest X-rays be presented: as a boundary case, or left out of the main text?

References

Noel Codella, Veronica Rotemberg, Philipp Tschandl, et al. Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (ISIC). *arXiv:1902.03368*, 2019.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020.

Haifan Gong, Guanqi Chen, Ranran Wang, Xiaoliang Xie, Mingzhi Mao, Yizhou Yu, Fei Chen, and Guanbin Li. Multi-task learning for thyroid nodule segmentation with thyroid region prior. In *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2021. doi: 10.1109/ISBI48211.2021.9434087.

Amelia Jiménez-Sánchez, Dovile Juodelyte, Bethany Chamberlain, and Veronika Cheplygina. Detecting shortcuts in medical images – a case study in chest x-rays. In *IEEE ISBI*, 2023.

Polina Kirichenko, Pavel Izmailov, and Andrew Gordon Wilson. Last layer re-training is sufficient for robustness to spurious correlations. In *ICLR*, 2023.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *TMLR*, 2024.

Fernando Pérez-García, Harshita Sharma, Sam Bond-Taylor, et al. Exploring scalable medical image encoders beyond text supervision. *Nature Machine Intelligence*, 2025.

Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M Summers. ChestX-ray8: Hospital-scale chest x-ray database and benchmarks. In *CVPR*, 2017.

Victoria Wegley et al. Dermatology skin lesion image analysis: artifact annotation masks for ISIC 2019. Scholars' Mine, Missouri S&T, doi:10.71674/man1-qa33, 2026.

Julia K Winkler, Christine Fink, Ferdinand Toberer, et al. Association between surgical skin markings in dermoscopic images and diagnostic performance of a deep learning convolutional neural network for melanoma recognition. *JAMA Dermatology*, 155(10):1135–1141, 2019.

Julia K Winkler, Katharina Sies, Christine Fink, et al. Association between different scale bars in dermoscopic images and diagnostic performance of a market-approved deep learning convolutional neural network for melanoma recognition. *European Journal of Cancer*, 145:146–154, 2021.

Siyuan Yan et al. Derm1M: A million-scale vision-language dataset aligned with clinical ontology knowledge for dermatology. In *ICCV*, 2025.

Akihito Yokote et al. The SEE-AI project dataset: capsule endoscopy images with lesion bounding boxes. Kaggle dataset capsuleyolo/kyucapsule, CC BY 4.0, 2023.

Qi Zhao, Shuchang Lyu, Wenpei Bai, Linghan Cai, Binghao Liu, Meijing Wu, Xiubo Sang, Min Yang, and Lijiang Chen. A multi-modality ovarian tumor ultrasound image dataset for unsupervised cross-domain semantic segmentation. *arXiv preprint arXiv:2207.06799*, 2022.